

David Gonzalez

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EDUCATION

University of Florida

Gainesville, FL

Bachelor of Science in Computer Science, Minor in Mathematics: GPA: 4.0

May 2028

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Databases and Information Systems.

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript

Developer Tools: Git, GDB, Linux OS, Docker, REST APIs, Github CI, Node.js, Vitest, Emscripten

Frameworks & Libraries: PyTorch, OpenCV, FastAPI, React, ROS, SvelteKit, Supabase, React, Tailwind

EXPERIENCE

Software Engineering Intern

May 2026 – Present

AvonRisk

Remote

- **AI Tooling & Evaluation:** Designed and built an empirical compatibility harness from scratch to evaluate AI-agent skills, combining deterministic outcomes with an LLM judge to validate compatibility and functionality predictions.
- **Testing & QA:** Designed comprehensive tests that increased coverage by up to 106%, most critically by implementing regression tests for bugs identified during QA.
- **Performance Optimization:** Lazy-loaded 93% of ESM modules and eliminated redundancy in git subprocess calls, accelerating command execution times by up to 85%.

Undergraduate Software Researcher

May 2026 – Present

Machine Intelligence Laboratory (MIL), University of Florida

Gainesville, FL

- **Autonomous Mission Planning:** Authored and tested XML mission plans within a ROS environment, directing the autonomous sub's robotic gripper to align with, grasp, and manipulate objects per mission objectives.
- **Sensor Integration & Navigation:** Designed software consuming camera, hydrophone, and onboard-sensor data from existing ROS subscriber nodes to orient the autonomous sub throughout each mission.

Teacher Assistant - Discrete Structures

January 2026 – May 2026

University of Florida

Gainesville, FL

- Mentored **55+** students through weekly discussion sections, reinforcing core discrete-math concepts.
- Hosted weekly office hours; section students scored **5% higher** than the overall course average on exams.
- Lesson-planned and authored over **33% of exam materials** to challenge students and assess understanding.

PROJECTS

Ledgr (Bank of America Code-a-thon Finalist) | *FastAPI, LiteLLM, Svelte 5, React Native, Bun* April 2026

- **Asynchronous Vision-LLM Pipeline:** Engineered a FastAPI backend utilizing LiteLLM to prompt Gemma-3 for structured JSON extraction of receipt data, implementing an EasyOCR and custom regex-based parser fallback.
- **Cross-Platform Monorepo Architecture:** Built a unified Bun and Turborepo workspace managing a SvelteKit web dashboard, a React Native mobile client, and a PostgreSQL database, deploying via Docker and Railway.

MyPose | *PyTorch, OpenCV, C++, FastAPI, Supabase, React*

March 2026 – Present

- **Machine Learning Model Development:** Architected a user-calibrated Siamese network on a PyTorch ST-GCN base, applying transfer learning to map individual kinematic manifolds for personalized biomechanical evaluation.
- **Real-Time API Integration:** Engineered an asynchronous FastAPI and MediaPipe computer-vision pipeline, extracting 33-node skeletal graphs into pgvector for fast spatial-temporal embedding search.
- **Data Analysis:** Built a C++ module with pybind11 and Dynamic Time Warping to temporally align sequences, computing joint-angle cosine similarities to isolate precise biomechanical deviations.

TraceAndPace | *C++, React, TypeScript, Emscripten, WebAssembly*

March 2026 – April 2026

- Engineered multithreaded backend to perform concurrent tree operations, ensuring validity of comparisons.
- Built a custom node condensation algorithm to reduce 1,000,000 data points into 50 recursive nodes.

EXTRACURRICULAR

ACM Design Team

April 2026 – Present

ACM

Gainesville, FL

- Built data-visualization components that surfaced engagement trends from historical attendance records, delivering insights for chapter leadership.
- Built a secure, location-based check-in system to automate real-time attendance tracking, replacing the chapter's manual sign-in process; launching in fall for **300+ members** across partnered clubs.